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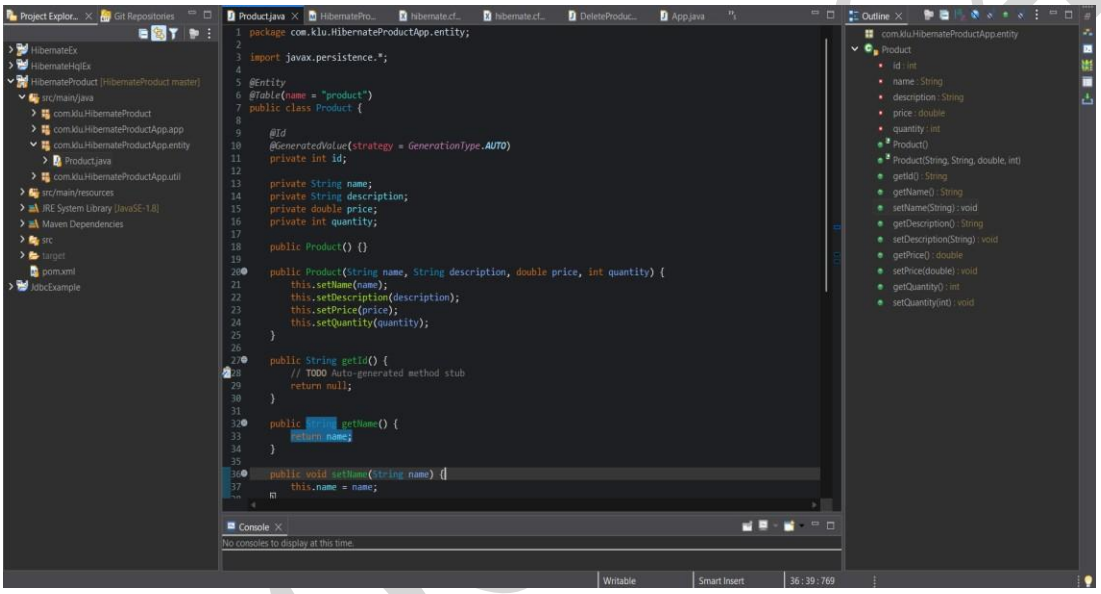
ID:	2400030645
SECTION:	S-24
SUBJECT:	FULL STACK APPLICATION DEVELOPMENT

#SKILL – 2 Hibernate CRUD Operations

Tasks:

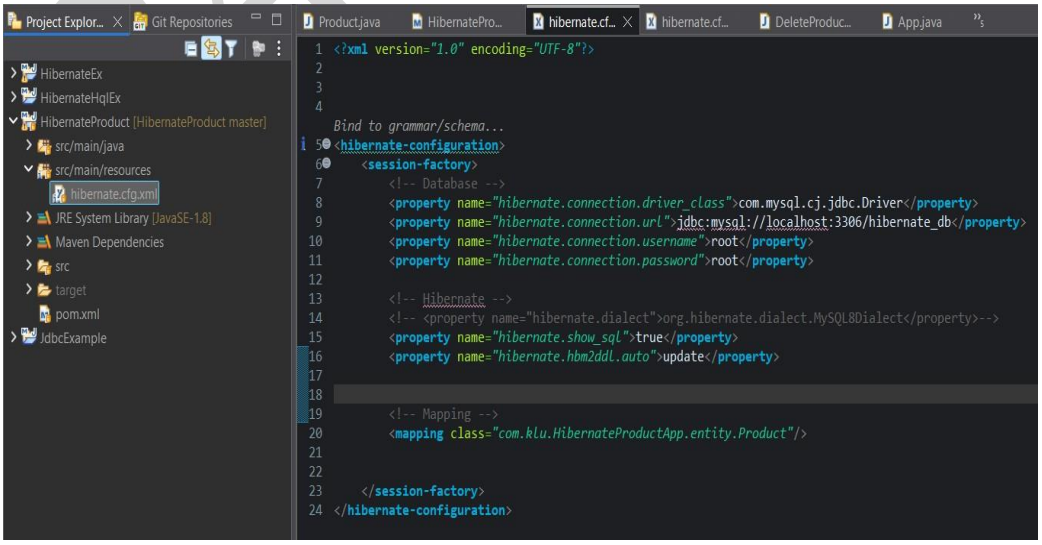
1. Create a Product entity with fields: id (primary key), name, description, price, quantity.

<u>Output:-</u>



2. Configure Hibernate and map the Product entity to a table using JPA annotation

<u>Output:-</u>



3. **Insert multiple Product records into the database.**

Output:-

```
package com.klu.HibernateProductApp.app;

import org.hibernate.Session;

public class InsertProduct {

    public static void main(String[] args) {

        Session session = HibernateUtil.getSessionFactory().openSession();
        Transaction tx = session.beginTransaction();

        Product p1 = new Product("Laptop", "HP", 60000, 5);
        Product p2 = new Product("Mobile", "Samsung", 30000, 10);

        session.save(p1);
        session.save(p2);

        tx.commit();
        session.close();

        System.out.println("Inserted Successfully");
    }
}
```

```
<terminated> InsertProduct [Java Application] C:\Users\sathv\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.v20251105-0741\jre\b
INFO: HHH000115: Hibernate connection pool size: 20 (min=1)
Jan 04, 2026 6:05:13 PM org.hibernate.dialect.Dialect <init>
INFO: HHH000400: Using dialect: org.hibernate.dialect.MySQL8Dialect
Jan 04, 2026 6:05:14 PM org.hibernate.resource.transaction.backend.jdbc.internal.DdlTransactionIsolatorNonJtaImpl getI
INFO: HHH10001501: Connection obtained from JdbcConnectionAccess [org.hibernate.engine.jdbc.env.internal.JdbcEnvironme
Jan 04, 2026 6:05:14 PM org.hibernate.engine.transaction.jta.platform.internal.JtaPlatformInitiator initiateService
INFO: HHH000490: Using JtaPlatform implementation: [org.hibernate.engine.transaction.jta.platform.internal.NoJtaPlatfo
Hibernate: select next_val as id_val from hibernate_sequence for update
Hibernate: update hibernate_sequence set next_val= ? where next_val=?
Hibernate: select next_val as id_val from hibernate_sequence for update
Hibernate: update hibernate_sequence set next_val= ? where next_val=?
Hibernate: insert into product (description, name, price, quantity, id) values (?, ?, ?, ?, ?)
Hibernate: insert into product (description, name, price, quantity, id) values (?, ?, ?, ?, ?)
Inserted Successfully
```

4. **Retrieve a product using its id.**

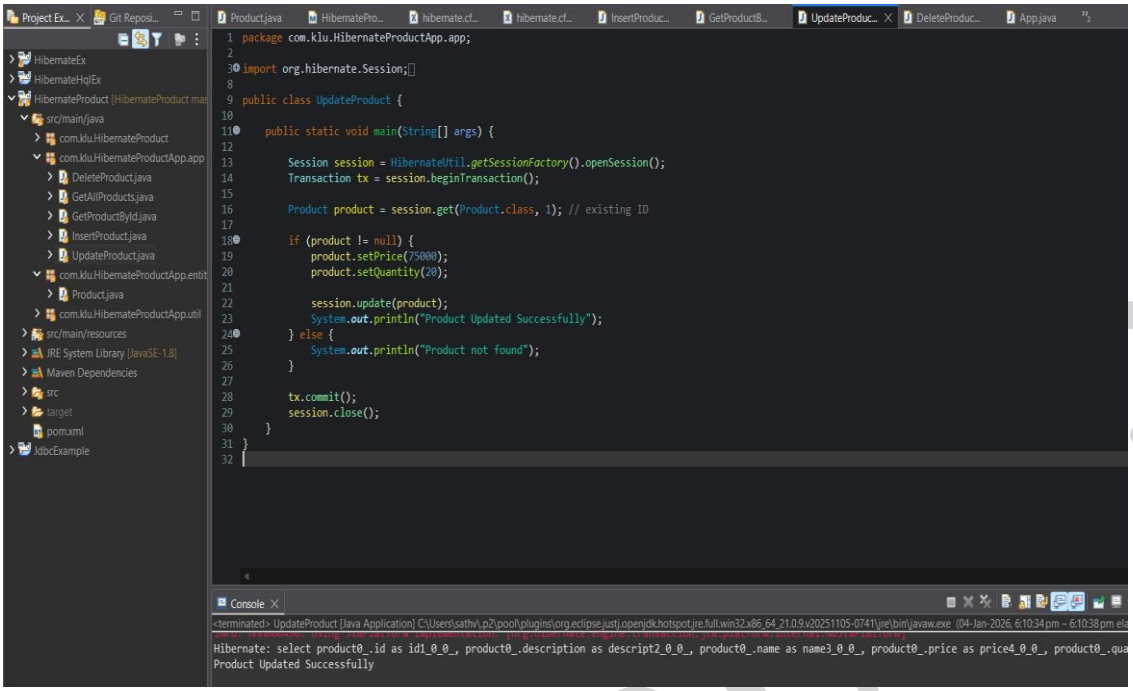
Output:-

```
package com.klu.HibernateProductApp.app;

import org.hibernate.Session;

public class GetProductById {

    public static void main(String[] args) {

        Session session = HibernateUtil.getSessionFactory().openSession();

        Product product = session.get(Product.class, 1); // change ID if needed

        if (product != null) {
            System.out.println("Product ID: " + product.getId());
            System.out.println("Name: " + product.getName());
            System.out.println("Price: " + product.getPrice());
            System.out.println("Quantity: " + product.getQuantity());
        } else {
            System.out.println("Product not found");
        }

        session.close();
    }
}
```

```
<terminated> GetProductById [Java Application] C:\Users\sathv\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.v20251105-0741\jre\bin\javaw.exe (04-Jan-2026, 6:08:22 pm – 6:08:26 pm elapsed 0:00:04
INFO: HHH000115: Hibernate connection pool size: 20 (min=1)
Jan 04, 2026 6:08:25 PM org.hibernate.dialect.Dialect <init>
INFO: HHH000400: Using dialect: org.hibernate.dialect.MySQL8Dialect
Jan 04, 2026 6:08:25 PM org.hibernate.resource.transaction.backend.jdbc.internal.DdlTransactionIsolatorNonJtaImpl getIsolatedConnection
INFO: HHH10001501: Connection obtained from JdbcConnectionAccess [org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator$ConnectionProviderJdbcConnectionAccess@
Jan 04, 2026 6:08:26 PM org.hibernate.engine.transaction.jta.platform.internal.JtaPlatformInitiator initiateService
INFO: HHH000490: Using JtaPlatform implementation: [org.hibernate.engine.transaction.jta.platform.internal.NoJtaPlatform]
Hibernate: select product0_.id as id1_0_0_, product0_.description as descript2_0_0_, product0_.name as name3_0_0_, product0_.price as price4_0_0_, product0_.quantity as c
Product ID: null
Name: Laptop
Price: 75000.0
Quantity: 20
```

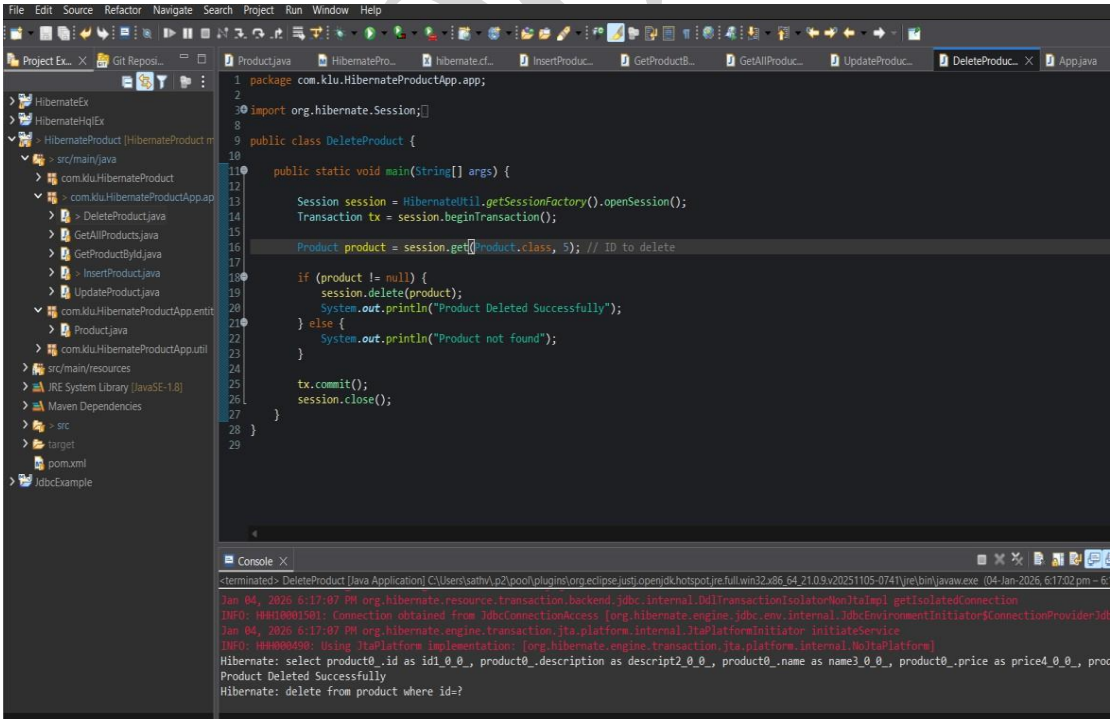
5. Update the price or quantity of any selected product.

Output:-



6. Delete a product record by id if it is discontinued

Output:-



7. a. Implement Hibernate ID generation strategies (AUTO, IDENTITY, SEQUENCE) inside the entity to observe how primary key values differ.

Output:-

```
<terminated> InsertProduct [Java Application] C:\Users\sathv\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64
Hibernate: select next_val as id_val from hibernate_sequence for update
Hibernate: update hibernate_sequence set next_val= ? where next_val=?
Hibernate: select next_val as id_val from hibernate_sequence for update
Hibernate: update hibernate_sequence set next_val= ? where next_val=?
Hibernate: select next_val as id_val from hibernate_sequence for update
Hibernate: update hibernate_sequence set next_val= ? where next_val=?
Hibernate: insert into product (description, name, price, quantity, id) values (?, ?, ?, ?, ?)
Hibernate: insert into product (description, name, price, quantity, id) values (?, ?, ?, ?, ?)
Hibernate: insert into product (description, name, price, quantity, id) values (?, ?, ?, ?, ?)
Inserted Successfully
```

SQL File 4*

```
select * from product
```

Result Grid

id	description	name	price	quantity
1	HP	Laptop	75000	20
3	HP	Laptop	60000	5
4	Samsung	Mobile	30000	10
6	Samsung	Mobile	30000	10
7	HP	Laptop	60000	5
8	Samsung	Mobile	30000	10
9	Apple	Tablet	45000	7
10	HP	Laptop	60000	5
11	Samsung	Mobile	30000	10
12	Apple	Tablet	45000	7
NULL	NULL	NULL	NULL	NULL

b. The ID can be manual or auto generated, based on your chosen strategy.

Navigator

SCHEMAS

- hibernate_db
  - Tables
    - hibernate_sequence
    - product
      - Columns
      - Indexes
      - Foreign Keys
      - Triggers
  - Views
  - Stored Procedures
  - Functions
- hibernatelab_db
- sample
- sys

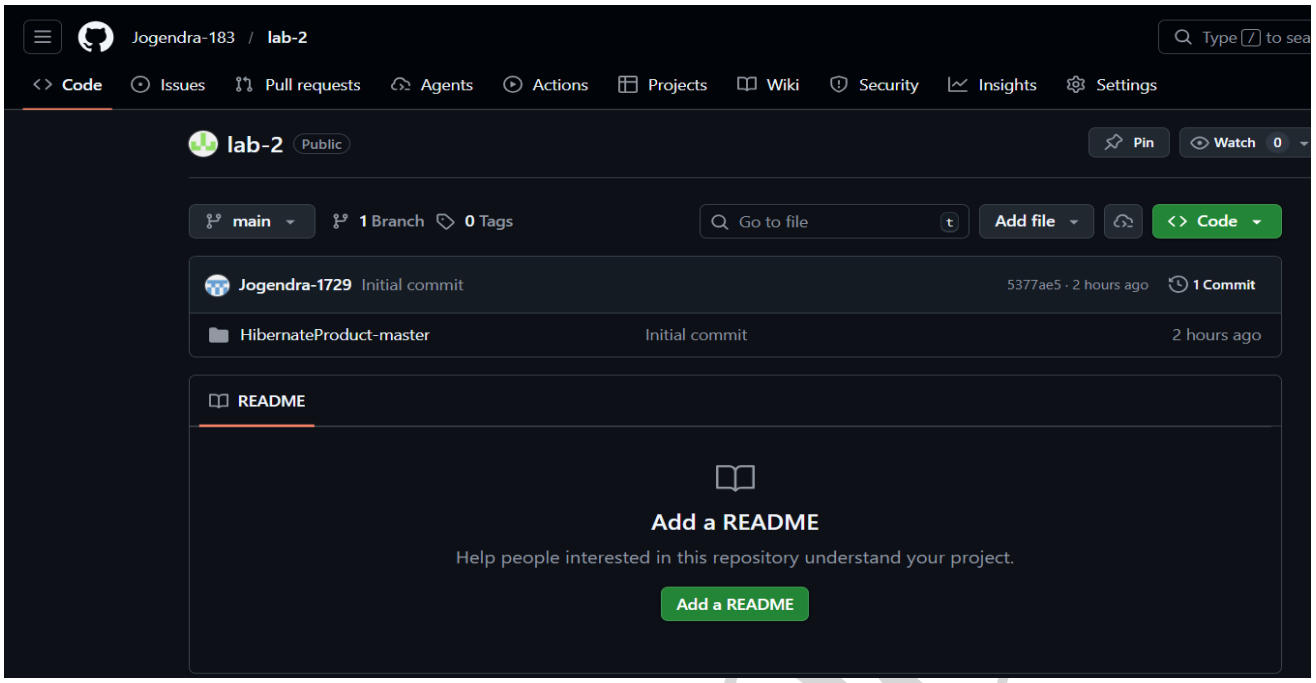
SQL File 4*

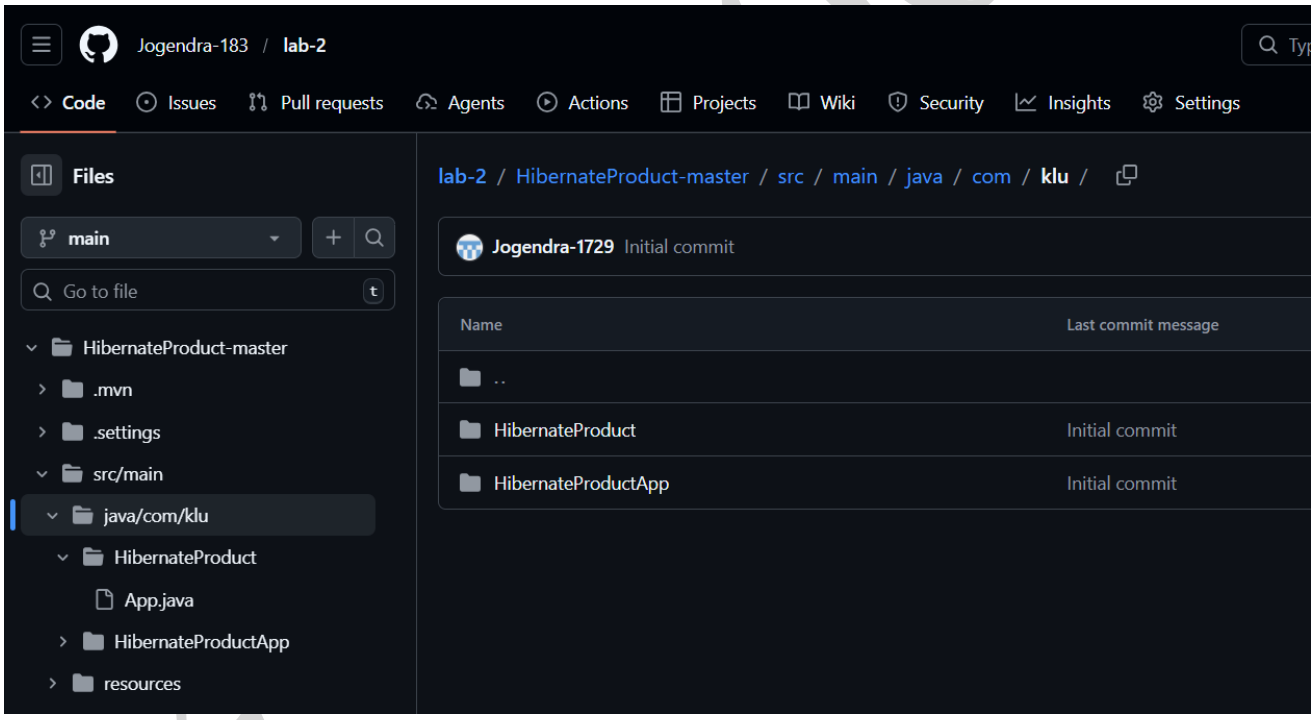
```
select * from hibernate_sequence
```

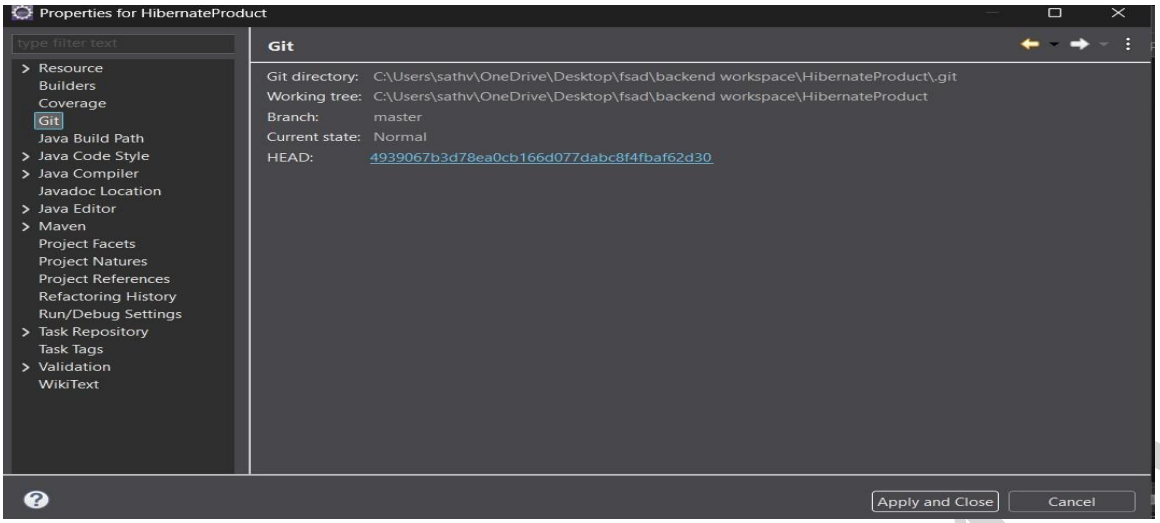
Result Grid

next_val
13

8. **Push the Hibernate project into a single GitHub repository**







Conclusion

This experiment successfully demonstrated the implementation of complete CRUD operations using Hibernate with JPA annotations for a retail inventory system. By mapping the Product entity to a database table and performing create, read, update, and delete operations, we validated how Hibernate simplifies database interaction through ORM. The use of different ID generation strategies (AUTO, IDENTITY, SEQUENCE) helped in understanding primary key management in real-world applications. Overall, this exercise strengthened practical knowledge of Hibernate configuration, entity mapping, and database operations, which are essential for building scalable Java-based enterprise applications.

GITHUB REPO LINK :- https://github.com/Jogendra-183/lab-2/tree/main

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